Toward Integrity and Detail With Ensemble Learning for Salient Object Detection in Optical Remote-Sensing Images

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***Abstract*— Optical remote-sensing image salient object detection (ORSI-SOD) poses significant challenges due to complicated object variances and interfering surroundings. Although existing methods have achieved impressive performance, they encounter difficulties in balancing deep and shallow features, leading to limitations in preserving object integrity and edge detail. To address this, we propose the integrated and detailed ensemble learning (IDEL) framework, which incorporates hierarchical branches with deep supervision. By divide-and-conquer, each branch captures information with a specific granularity, while the fusion module combines all outputs to generate the final saliency maps. To ensure the effectiveness of ensemble learning, IDEL is designed to satisfy two necessary conditions: the weak learner property and branch independence. Firstly, we utilize the Transformer blocks with a global receptive field and purify intermediate features with the deep supervision module (DSM) to enhance the performance of each branch. Secondly, we disentangle multiple branches through hardness-aware weights and hierarchical supervision labels, allowing them to learn distinct features. Qualitative visualizations demonstrate the effectiveness of each module, and extensive experimental results conducted on three popular ORSI datasets confirm the superiority of IDEL compared to other state-of-the-art (SOTA) counterparts.**

***Index Terms*— Deep supervision, ensemble learning, optical remote-sensing images (ORSIs), salient object detection (SOD), transformer.**

NOMENCLATURE

i	Branch index.
f^i	Encoder in each branch.
g^i	Deep supervision module in each branch.
h^i	Decoder in each branch.
z_e^i	Output feature of encoder.
z_{ca}^i	Feature after channel attention.
z_{fg}^i	Foreground feature.
z_{ds}^i	Output feature of deep supervision module.
z_{dec}^i	Output feature of decoder.
$\boldsymbol{y}$	Original label.
$\boldsymbol{y}_{\text{dt}}$	Detail part of deep supervision label.
$\boldsymbol{y}_{\text{bd}}$	Body part of deep supervision label.
$\hat{\boldsymbol{y}}^i$	Saliency map of each branch.
$\hat{\boldsymbol{y}}$	Final saliency map.

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I. INTRODUCTION

SALIENT object detection (SOD) aims to identify visually prominent objects and generates the pixel-level binary saliency maps [1]. Initially developed for natural scene images (NSIs) with great success, SOD has been extended to various types of images (e.g., RGB-D images [2], RGB-T images [3], and video [4]) and found wide applications in downstream vision tasks (e.g., object segmentation [5], object tracking [6], image quality assessment [7], and image retargeting [8]). Inspired by techniques of SOD in NSIs, SOD in optical remote-sensing images has experienced rapid development recently and facilitated numerous tasks including image fusion [9], scene classification [10], object extraction [11], and oil tank detection [12]. The advancement of deep learning techniques has significantly facilitated the evolution of NSI-SOD. In recent years, researchers have introduced several advanced methods [13], [14], [15], [16], [17], [18], [19] based on fully convolutional neural networks (FCNs). However, ORSIs possess unique characteristics, including multiscale objects, variable number of objects, messy object distribution, and cluttered background. Consequently, high-performance methods from NSIs are not applicable to ORSIs.

To address the aforementioned challenges, most of the advanced optical remote-sensing image SOD (ORSI-SOD) works design subtle modules and frameworks [21], [22], [23], [24], [25], [26], [27], [28] based on the characteristics of ORSIs to generate complementary information with multiscale features. However, we observed that existing methods still suffer from incomplete objects and unclear edges and struggle to strike a balance between integrity and detail, as depicted in Fig. 1(a). Previous works [29] reveal that deep features tend to prioritize capturing integrated information, whereas shallow features are inclined to details. Hence, we apply the divide-and-conquer approach to effectively learn features from different layers, as illustrated in Fig. 1(b). Specifically,

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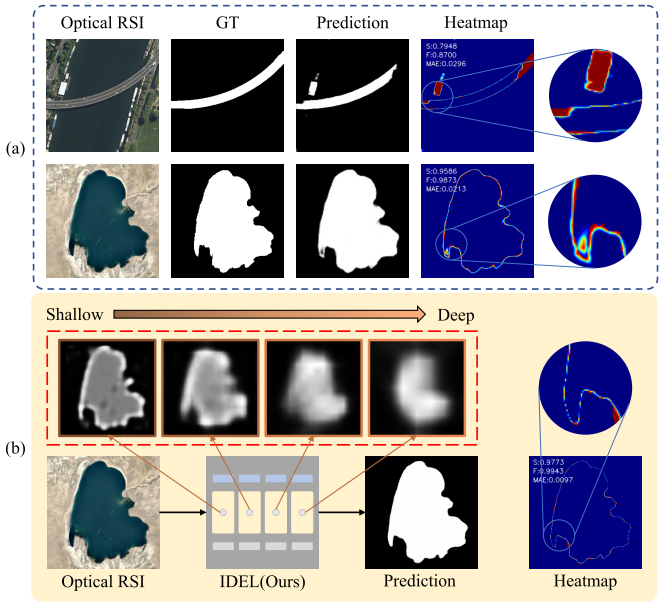

Fig. 1. (a) We present error pixel heatmaps and corresponding evaluation metrics for selected prediction samples from existing methods. In the heatmaps, red color indicates a high pixel error, while blue color indicates a low pixel error. The first row showcases results from ASTT [20], and the second row displays results from DAFNet [21]. (b) Our proposed IDEL framework addresses the challenge of simultaneously ensuring object integrity and edge detail. The visualization of intermediate features serves as a clear representation of the divide-and-conquer nature of our method.

we adopt a pyramidal structure in which each branch comprises an encoder, a decoder, and a deep supervision module (DSM). For parametric efficiency, the deep branch shares the encoder and decoder of the shallow branch and sequentially stacks additional blocks at the end.

In the ensemble learning paradigm, it is expected that different branches will acquire features of various granularity, which will be subsequently fused to compute the final saliency maps. Therefore, it is imperative for all branches to fulfill two necessary conditions: 1) possessing the weak learner property individually and 2) maintaining independence among branches. Firstly, to ensure that each branch functions as a weak learner, it is necessary to build each branch employing modules with strong learning capabilities. We select the transformer modules as encoders due to their global receptive field in a single layer, which is hardly achievable with convolutional neural networks (CNNs). Additionally, there have been limited endeavors [20] to incorporate the vision transformers (ViTs) [30] into ORSI-SOD. Specifically, we choose pyramid pooling transformer (P2T) [31] as our encoder because it reduces sequence length while capturing powerful contextual features. Moreover, we deploy the DSM to each branch, drawing inspiration from multicontent complementation module (MCCM) [24], which effectively purifies the intermediate features of each branch to obtain the foreground features. In DSM, we impose constraints on the foreground features using deep supervision labels, which effectively alleviates the multiple-solution issue in the optimization process due to insufficient training data from ORSIs. Deep supervision has proven its effectiveness in several methods [32], [33]. Some prior ORSI-SOD methods [21], [23], [24] have employed deep supervision, but they solely rely on edge labels for this purpose. These approaches are not justified as the network exhibits varying inductive biases with depth. Shallow branches specialize in detecting subtle patterns while deep branches are better equipped to capture holistic patterns. To cater to the inductive biases of different branches, we elaborately decouple labels into two parts, namely detail and body, and assign weights to each part based on the corresponding depth. The hierarchical deep supervision labels are obtained by taking the weighted sum of the detail and body labels. Secondly, it is essential to ensure the independence of multiple branches in order to enhance the overall fused performance in ensemble learning. Our hierarchical deep supervision labels assign distinct training signals to different branches to promote independence. However, relying solely on subjective assignments is inadequate for accommodating the flexible learning process. Inspired by Adaboost-R2 algorithm [34], we propose the hardness-aware pixel-wise weighting strategy to dynamically adapt the learning state of the model. Pixels with a high error rate in each branch will be assigned large weights in the subsequent branch so that the errors will be amended recursively. These weights compel different branches to focus on distinct pixels, thereby enhancing branch diversity in conjunction with the hierarchical deep supervision labels.

In summary, we propose the integrated and detailed ensemble learning (IDEL) framework for ORSI-SOD. This framework utilizes a hierarchical network with multiple branches to achieve multilevel granularity. It accurately predicts saliency maps by aggregating the outputs of multiple branches. The experimental results, both qualitative and quantitative, conducted on three widely used ORSI datasets, demonstrate that the IDEL effectively captures both holistic and subtle features while achieving superior performance. Our main contributions are summarized as follows.

1) *An effective ensemble framework, IDEL:* Our proposed IDEL framework effectively captures features with multigranularity simultaneously by leveraging multiple branches, which are fused to generate final saliency maps.
2) *Introduction of global-receptive-field modules:* We employ a transformer encoder with global receptive field, specifically P2T, and incorporate DSMs to enhance feature learning within each branch.
3) *Hardness-aware pixel-wise weighting:* We design a hardness-aware weighting mechanism that enables different branches to adaptively focus on distinct pixels, thereby improving the independence among branches.
4) *Consistent improvement of ORSI-SOD task:* Our method outperforms 28 state-of-the-art (SOTA) SOD methods in extensive qualitative and quantitative experiments conducted on three ORSI datasets, and ablation experiments confirm the validity of our proposed components.

The rest of this article is organized as follows. Section II provides a review of related work on NSI-SOD and ORSI-SOD. Section III presents the details of our proposed method. Section IV reports the comprehensive evaluation and ablation experiments. Section V concludes the article.

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II. Related Work

A. SOD in NSIs

Traditional SOD models extracted handcrafted features based on simulated human attention mechanisms [35]. They relied on heuristic cues, such as contrast [36], color space [37], center prior [38], and boundary prior [39]. Itti et al. [40] proposed the pioneering image-based visual saliency model, setting a precedent for subsequent research. Zhai and Shah [41] defined pixel-level saliency by comparing it to the luminance information of all other pixels. Although these traditional methods performed well in simple scenarios, they are limited by the lack of exploitation of advanced semantic features in complex scenarios.

With the remarkable advancements in deep learning, NSI-SOD has achieved impressive performance by leveraging CNNs due to their strong expressive power. Pang et al. [13] proposed an aggregated interaction module to integrate features from adjacent layers and utilized a self-interaction module at the decoder to effectively utilize multiscale features. Zhao et al. [14] introduced a multiscale gate unit to optimize the transfer of valuable contextual information from the encoder to the decoder. Zhuge et al. [16] developed complete cognitive networks to address the integrity of salient objects. Some researchers have employed well-designed labels, in addition to the original masks, for supervision [17], [19]. Wei et al. [17] proposed assigning higher prediction difficulty to edges and their nearby pixels, and divided the labels into detail and body maps to supervise the network. Yang et al. [19] constructed connectivity masks as additional labels to effectively model the relationships between edges and their nearby pixels. With the advent of ViT [30], the high-performing transformer has garnered attention. Liu et al. [15] predicted saliency solely based on transformers, capturing long-range dependencies that CNNs lack. Yun and Lin [42] designed the self-refined transformer network, incorporating a global context branch and a context refinement module (CRM).

While existing NSI-SOD methods have demonstrated good performance, directly applying these methods to ORSIs is not feasible. However, this limitation does not prevent us from seeking inspiration from these NSI-SOD methods.

B. SOD in Optical ORSIs

ORSI-SOD has received less attention compared to NSI-SOD, despite its significance. Earlier ORSI-SOD methods [43], [44], [45] relied on handcrafted features, resulting in limited performance. However, with the increasing popularity of deep learning, more powerful methods have emerged. Li et al. [22] introduced the first publicly available ORSSD dataset and a two-stream pyramid module to address the multiscale and detailed challenges in SOD. ORSI-SOD has gained widespread attention, leading to the development of larger datasets such as EORSSD [21] and ORI4199 [25], as well as cleverly designed methods. Zhang et al. [21] proposed the delicate dense attention fluid (DAF) structure, which utilizes low-level attention cues to guide the generation of high-level attention maps. Zhou et al. [23] extracted multiscale features from three different resolutions of the input and explicitly and implicitly incorporated salient edge cues. Li et al. [24] designed the MCCM to extract multiple complementary information, including foreground features, background features, edge features, and global image-level features, to better highlight salient objects through attention mechanisms. Li et al. [26] focused on exploring the relationship and coordination of adjacent features in encoder–decoder networks. Tu et al. [25] extracted multiscale regional features and edge features, and employed two decoders to process and aggregate the features, distinguishing their approach from other methods. Wang et al. [27] replaced the last two layers of the commonly used CNN encoder with a transformer structure to create a new hybrid encoder. Liu et al. [28] designed the global perception module (GPM) to capture the relationships of the entire dataset from a global perspective and incorporated the module into the encoder. Gao et al. [20] designed an adaptive spatial tokenization transformer encoder to balance the transformer's ability to learn long-range relationships and its applicability to OSRI-SOD. Additionally, several methods [21], [23], [24], [27] utilized extra salient edge masks for supervision and employed hybrid loss functions.

The aforementioned methods effectively utilize multiscale features by incorporating well-engineered modules and networks to extract multiple complementary information, which is crucial in ORSI-SOD. However, they face limitations in capturing features with distinct granularity simultaneously, which hinders their ability to ensure object integrity and edge detail in challenging scenarios. In contrast, our method adopts the concept of ensemble learning to divide and conquer features across different granularity. This approach enables our method to surpass existing methods in performance.

III. Method

In this section, we provide a detailed presentation of our proposed IDEL framework. We first overview our framework in Section III-A, which mainly consists of four branches, each comprising an encoder, a decoder, and a DSM. Subsequently, we elaborate on our transformer encoder and DSM, which are crafted to establish each branch as a weak learner in Section III-B, Moreover, we introduce the hardness-aware pixel-wise weighting strategy and the fusion mechanism of all branches, which are employed to ensure branch independence in Section III-C. Finally, we present the optimization objectives and training details in Section III-D.

A. Overview: Ensemble Learning Framework

To start with, we define the input image sample as $\boldsymbol{x} \in \mathcal{D}$ ($\mathcal{D}$ is the training dataset) and the corresponding saliency map as $\boldsymbol{y} \in [0,1]^{h \times w}$, where h and w are the height and width, respectively. Our ultimate objective is to establish a learnable model $\mathcal{T}$ that effectively predicts the precise saliency map $\hat{\boldsymbol{y}} = \mathcal{T}(\boldsymbol{x})$. Considering the intricate object variances and interfering surroundings in ORSI-SOD, we explore to capture features at multiple granularities with multiple branches in an ensemble learning manner. Each branch consists of an encoder,

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a decoder, and a DSM, as illustrated in Fig. 2. Specifically, we denote the encoder in each branch as f^i and the output feature as $z_e^i \in \mathbb{R}^{c_i \times h_i \times w_i}$, where $i \in \{1, \ldots, N\}$[1] is the branch index and (c_i, h_i, w_i) indicates the shape of output feature z_e^i. Subsequently, we project the output of each encoder z_e^i to the foreground feature z_{fg}^i and the deep supervised feature z_{ds}^i by the DSM g^i. To construct the deep supervision label y_{ds}^i for each branch, we decouple the ground-truth (GT) label y into detail part y_{dt} and body part y_{bd} according to the inductive bias of the corresponding branch. Finally, we employ the decoder h^i to predict the saliency map $\hat{y}^i$ of each branch and fuse them to obtain the final saliency map $\hat{y}$. The notations we use are described in Nomenclature. In general, our framework IDEL constructs a nested bottleneck network. We will present a comprehensive depiction of each component in Sections III-B and III-C.

B. Condition 1: Weak Learner Property

A prerequisite for implementing ensemble learning is that each branch functions as a weak learner at minimum. To ensure this, we primarily focus on the expressive capabilities and supervision methods of each branch. Regarding expressive capability, we primarily consider the choice of encoders. For the supervision method, we design the DSMs with elaborate supervision labels.

1) Encoder Choice: Shallow convolutional networks face difficulty in capturing global semantic information due to their limited receptive field. Hence, we adopt P2T [31] blocks as the encoder f^i due to its global receptive field in both deep and shallow features extraction. To reduce the sequence length and capture powerful contextual features, P2T introduced pyramid pooling to multihead self-attention in the vision transformer. We resize the input image x to 224×224, and after four stages we can get a set of sequential features as $\{z_e^i | i = 1, \ldots, N\}$,[2] which is computed as follows:

$$z_e^i = \begin{cases} f^i\left(z_e^{i-1}\right), & i = 1, \ldots, N \\ x, & i = 0 \end{cases} \tag{1}$$

where N denotes the number of branches.

2) Deep Supervision Module: Deep supervision provides valuable guidance for learning intermediate features, which is advantageous compared to the conventional end-to-end approach. Recognizing that the features extracted from P2T may be relatively coarse, we introduced the DSM to further extract foreground features and apply deep supervision to enhance their quality. The design of the DSM is inspired by the MCCM [24], which incorporates channel attention [46] and spatial attention [24]. In Fig. 3, we firstly apply channel attention to the features z_e^i from P2T to reduce redundant information. The process is illustrated as follows:

$$z_{\text{ca}}^i = \mathcal{A}_c\left(z_e^i\right) \odot z_e^i \tag{2}$$

where z_{ca}^i denotes the streamlined feature, $\mathcal{A}_c$ means channel attention, and $\odot$ represents channel-wise multiplication. Subsequently, we can get the foreground feature z_{fg}^i by spatial attention, which is described as

$$z_{\text{fg}}^i = \mathcal{A}_s\left(z_{\text{ca}}^i\right) \tag{3}$$

where $\mathcal{A}_s$ indicates spatial attention, and the foreground feature z_{fg}^i will be imposed with deep supervision. Finally, we can obtain the deep supervised feature z_{ds}^i as

$$z_{\text{ds}}^i = z_{\text{ca}}^i \odot \left(1 + z_{\text{fg}}^i\right). \tag{4}$$

3) Deep Supervision Labels: The choice of deep supervision labels plays a crucial role in guiding the learning of intermediate features. Previous approaches [21], [23], [24] commonly utilize salient object edges as deep supervision labels, but they may not align with the network's inductive bias. In our approach, we refine the deep supervision labels to suit the inductive biases of different branches better.

To achieve this, we decouple the original label into two parts: the detail part that contains the edge information, and the body part that represents the body regions, and we construct the deep supervision label in a weighted sum manner as follows:

$$\left(y_{\text{dt}}, y_{\text{bd}}\right) \xleftarrow{\text{decouple}} y \tag{5}$$

$$y_{\text{ds}}^i = y_{\text{dt}} \times \alpha_i + y_{\text{bd}} \times (1 - \alpha_i) \tag{6}$$

where y is the original label, and y_{ds}^i refers to the deep supervision label of the corresponding branch. This approach acknowledges that salient object edges and the main body have different learning difficulties and granularity. By adjusting the hyperparameter α_i to manipulate the weighted sum, we can better cater to the inductive bias of different branches. Initially, we attempt to transform these hyperparameters into learnable variables. However, during the optimization process, these parameters fluctuate with each batch update, leading to instability that impacts the model's performance. Consequently, we opt for manual adjustments. In particular, we set $\{\alpha_1, \alpha_4\}$ to $\{1, 0\}$ to enable the shallowest and deepest layers to focus on detail features and holistic features, respectively. For intermediate layers, they concern the mixed features between detail features and holistic features, so we let $\{\alpha_2, \alpha_3\}$ vary between 0 and 1 depending on the dataset. Fig. 2 illustrates two label decoupling methods. For shallow branches, we separate labels into edge pixels y_{dt}^1 and remaining pixels $y_{\text{bd}}^1 = y - y_{\text{dt}}^1$, ensuring clear boundaries to capture fine-grained features. Specifically, y_{dt}^1 is obtained as follows:

$$y_{\text{dt}}^1 = \mathcal{P}_e(y) \tag{7}$$

where $\mathcal{P}_e$ is the same salient object edge generation process as in EGNet [47]. For deep branches, we employ a soft split of labels using distance transformations (DTs) [17] into edges and their neighboring pixels y_{dt}^2, as well as body pixels y_{bd}^2, to prioritize holistic features. Specifically, the input of DT is the original label y, which can be divided into two groups: foreground y_{fg} and background y_{bg}. For each pixel p, the corresponding pixel value is 1 if it belongs to the foreground

[1] N represents the total number of branches, and N is equal to 4 in our method.

[2] Specifically, N is equal to 4, and the corresponding features have dimensions of 56×56, 28×28, 14×14, and 7×7.

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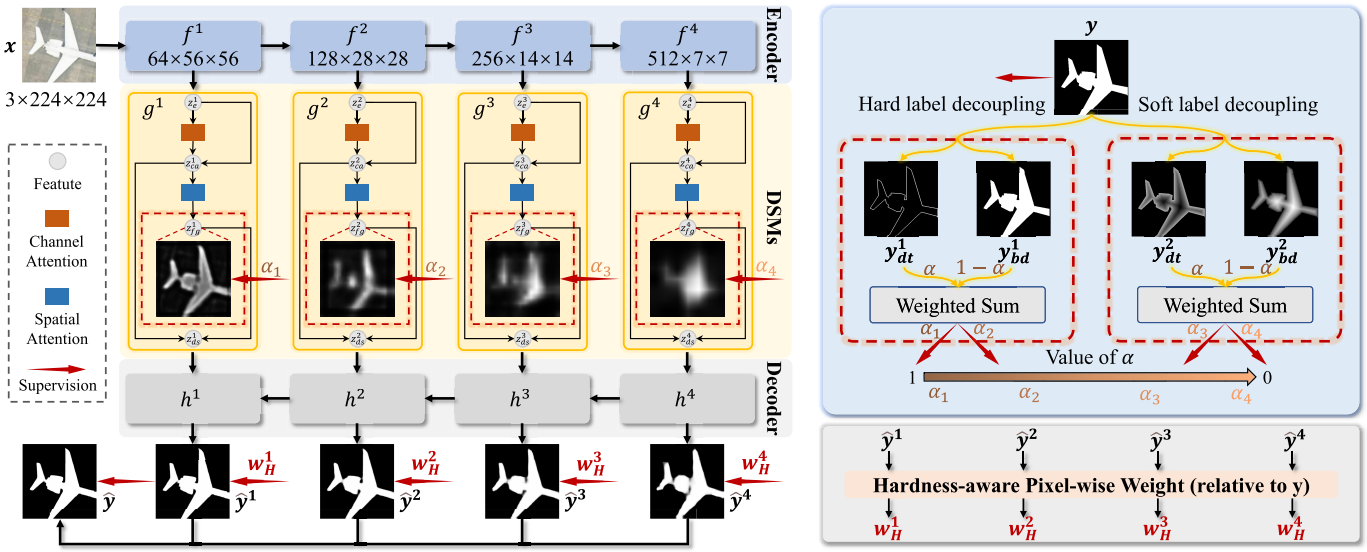

Fig. 2. IDEL framework consists of four branches, labeled as the ith branch ($i \in \{1, 2, 3, 4\}$) from left to right. Each branch consists of an encoder block, a DSM, and a decoder block. The outputs of the four branches are fused together to generate the final saliency map. Additionally, we provide detailed information about hierarchical deep supervision labels and hardness-aware pixel-wise weights.

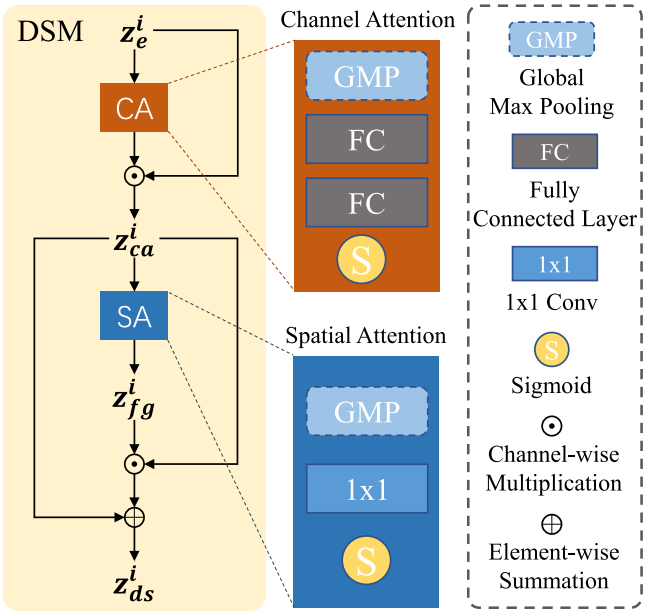

Fig. 3. Details of the structure of DSM.

and 0 if it belongs to the background. To acquire the DT result of $\boldsymbol{y}$, a metric function $d(p, q) = ((p_x - q_x)^2 + (p_y - q_y)^2)^{1/2}$ is defined to measure the distance between pixels. If pixel p belongs to the foreground, DT initially locates its nearest pixel q in the background and calculates the distance using $d(p, q)$. If pixel p belongs to the background, its minimum distance is set to zero. Consequently, the distance transformation can be represented as follows:

$$\boldsymbol{y}'(p) = \begin{cases} \min\limits_{q \in \boldsymbol{y}_{\text{bg}}} d(p, q), & p \in \boldsymbol{y}_{\text{fg}} \\ 0, & p \in \boldsymbol{y}_{\text{bg}}. \end{cases} \tag{8}$$

After the distance transformation, we normalize the pixel values in $\boldsymbol{y}'$ to $[0, 1]$. At this point, the pixel in $\boldsymbol{y}'$ that lies within the center of the salient object possesses a higher value, signifying that $\boldsymbol{y}'$ represents the body portion of the original label $\boldsymbol{y}$. As such, we can obtain the detail portion by subtracting the body portion $\boldsymbol{y}'$ from the original label $\boldsymbol{y}$. Ultimately, the soft label decoupling results in the body part label $\boldsymbol{y}^2_{\text{bd}}$ and detail part label $\boldsymbol{y}^2_{\text{dt}}$ as follows:

$$\begin{cases} \boldsymbol{y}^2_{\text{bd}} = \boldsymbol{y} \cdot \boldsymbol{y}' \\ \boldsymbol{y}^2_{\text{dt}} = \boldsymbol{y} \cdot (1 - \boldsymbol{y}'). \end{cases} \tag{9}$$

More details about deep supervision labels will be explained in Sections IV-C1 and IV-C6.

Algorithm 1 Calculate Hardness-Aware Pixel-Wise Weight

Input: $\boldsymbol{y} = \{j, y(j)\}_{j=1}^{h \times w}$, $\hat{\boldsymbol{y}}^i = \{j, \hat{y}^i(j)\}_{j=1}^{h \times w}$.
Output: Difficulty Weight $\boldsymbol{w}^i_H$.

Initialize $\boldsymbol{w}^i(j) = 1, j = 1, \ldots, h \times w$;
Calculate the maximum pixel error $E = max|\hat{y}^i(j) - y(j)|$;
for $j = 1, \ldots, h \times w$ **do**
  Calculate the relative pixel error $e(j) = 1 - exp(\frac{-|\hat{y}^i(j) - y(j)|}{E})$;
Calculate the error rate $\varepsilon = \sum_{j=1}^{h \times w} \frac{\boldsymbol{w}^i(j) e(j)}{hw}$;
Set the weighting factor to be $\beta = \frac{\varepsilon}{1-\varepsilon}$;
Set the difficulty weight to be $\boldsymbol{w}^i_H(j) = \boldsymbol{w}^i(j) \beta^{1-e(j)}$;
Normalization $\boldsymbol{w}^i_H(j) = \frac{\boldsymbol{w}^i_H(j)}{Z} hw$, $Z = \sum_{j=1}^{h \times w} \boldsymbol{w}^i_H(j)$;
Return $\boldsymbol{w}^i_H$.

C. Condition 2: Independency Among Branches

The other prerequisite for implementing ensemble learning is to maintain the independence of multiple branches. While our hierarchical deep supervision labels contribute to promoting independence by assigning distinct supervised signals to different branches, subjective assignment alone may not be sufficient for a flexible learning process. To address this,

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we emphasize on dynamically regulating the learning state of different branches. Inspired by Adaboost-R2 algorithm [34], we propose a hardness-aware pixel-wise weighting strategy.

1) Decoder Structure: We decode the features z_{ds}^i obtained by DSM through a decoder block to yield the saliency map $\hat{\boldsymbol{y}}^i$ of different branches. The features z_{ds}^i are first passed through three Conv-BN-ReLU layers, and then, respectively, through a ConvTransposed-BN-ReLU layer and a Conv-BN-ReLU layer, to obtain the decoded feature z_{dec}^i and saliency maps $\hat{\boldsymbol{y}}^i$ accordingly, which is formulated as

$$\left[\hat{\boldsymbol{y}}^i, z_{\text{dec}}^i\right] = \begin{cases} h^i\left(z_{\text{ds}}^i \copyright z_{\text{dec}}^{i+1}\right), & i = 1, \ldots, N-1 \\ h^i\left(z_{\text{dec}}^{i+1}\right), & i = N \end{cases} \tag{10}$$

where h^i indicates decoder block, © refers to concatenate, and N means the number of decoding blocks.

2) Hardness-Aware Pixel-Wise Weighting and Fusion: To dynamically adapt the learning state of the model, we calculate the hardness-aware pixel-wise weight $\boldsymbol{w}_H^i$ for each branch's saliency map $\hat{\boldsymbol{y}}^i$ in comparison to the original label $\boldsymbol{y}$. Each weight $\boldsymbol{w}_H^i$ is then assigned to the corresponding branch for computing the loss function, which serves the dual purpose of focusing on challenging pixels and maintaining the independence of the branches. Concretely, we recurrently calculate the error of the corresponding pixels in $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}^i$, where $\hat{\boldsymbol{y}}^i$ is resized to the same size as $\boldsymbol{y}$. The pixel weights are positively correlated with the pixel errors, as outlined in Algorithm 1, which are depicted as follows:

$$\boldsymbol{w}_H^i(j) = \mathcal{W}_H\left(\left|\boldsymbol{y}(j) - \hat{\boldsymbol{y}}^i(j)\right|\right) \tag{11}$$

where $\mathcal{W}_H$ means the weighting operator and $j \in \{1, \ldots, h \times w\}$ indicates the pixel indexes. Then, we fuse the saliency maps $\hat{\boldsymbol{y}}^i$ of different branches to obtain the final saliency map $\hat{\boldsymbol{y}}$, and the process is shown as follows:

$$\hat{\boldsymbol{y}} = \mathcal{C}\left(\hat{\boldsymbol{y}}^1 \copyright \hat{\boldsymbol{y}}^2 \copyright \cdots \copyright \hat{\boldsymbol{y}}^N\right) \tag{12}$$

where $\mathcal{C}$ is a single-layer 1×1 convolution, and © is concatenation.

D. Optimization Objective and Training Method

When computing the loss function, we ensure that the input is scaled to the same size as the target. Initially, we utilize the mean-square error (MSE) loss to facilitate the learning of foreground features z_{fg}^i that benefit from deep supervision

$$\mathcal{L}_{\text{fg}} = \sum_{i=1}^{N} \mathcal{L}_{\text{MSE}}\left(z_{\text{fg}}^i, \boldsymbol{y}_{\text{ds}}^i\right). \tag{13}$$

$\mathcal{L}_{\text{MSE}}$ is calculated as follows:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{h \times w} \sum_{j=1}^{h \times w} \left(z_{\text{fg}}^i(j) - \boldsymbol{y}_{\text{ds}}^i(j)\right)^2 \tag{14}$$

where $z_{\text{fg}}^i(j)$ denotes the jth pixel of the foreground prediction result, $\boldsymbol{y}_{\text{ds}}^i(j)$ represents the jth pixel of the deep supervision label, and h, w indicate the size of the input. Subsequently, we adopt a hybrid loss function including binary cross-entropy (BCE) loss, intersection over union (IoU) loss, and F-m loss [48] to further enhance the quality of the saliency map $\hat{\boldsymbol{y}}^i$ of different branches and the final saliency map $\hat{\boldsymbol{y}}$. The BCE loss, IoU loss, and F-m loss of the overall process can be expressed as follows:

$$\mathcal{L}_{\text{bce}} = \sum_{i=1}^{N} \mathcal{L}_{\text{BCE}}^{\boldsymbol{w}_H^i}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) + \mathcal{L}_{\text{BCE}}\left(\hat{\boldsymbol{y}}, \boldsymbol{y}\right) \tag{15}$$

where $\mathcal{L}_{\text{BCE}}^{\boldsymbol{w}_H^i}$ denotes the weighted BCE loss, and $\boldsymbol{w}_H^i$ refers to the previously obtained hardness-aware weight

$$\mathcal{L}_{\text{iou}} = \sum_{i=1}^{N} \mathcal{L}_{\text{IoU}}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) + \mathcal{L}_{\text{IoU}}\left(\hat{\boldsymbol{y}}, \boldsymbol{y}\right) \tag{16}$$

$$\mathcal{L}_{\text{fm}} = \sum_{i=1}^{N} \mathcal{L}_{\text{Fm}}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) + \mathcal{L}_{\text{Fm}}\left(\hat{\boldsymbol{y}}, \boldsymbol{y}\right). \tag{17}$$

Concretely, $\mathcal{L}_{\text{BCE}}$ is calculated as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{h \times w} \sum_{j=1}^{h \times w} \left[\boldsymbol{y}(j) \log\left(\hat{\boldsymbol{y}}^i(j)\right) + (1 - \boldsymbol{y}(j)) \log\left(1 - \hat{\boldsymbol{y}}^i(j)\right)\right] \tag{18}$$

where $\boldsymbol{y}(j)$ denotes the jth pixel of the GT, and $\hat{\boldsymbol{y}}^i(j)$ represents the jth pixel of the saliency predicted result. $\mathcal{L}_{\text{IoU}}$ can be expressed by the formula as

$$\mathcal{L}_{\text{IoU}} = 1 - \frac{\sum_{j=1}^{h \times w} \hat{\boldsymbol{y}}^i(j) \cdot \boldsymbol{y}(j)}{\sum_{j=1}^{h \times w} \left[\hat{\boldsymbol{y}}^i(j) + \boldsymbol{y}(j) - \hat{\boldsymbol{y}}^i(j) \cdot \boldsymbol{y}(j)\right]}. \tag{19}$$

The details of the calculation of $\mathcal{L}_{\text{Fm}}$ are as follows:

$$\mathcal{L}_{\text{Fm}} = 1 - \frac{(1 + \beta) \cdot P\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) \cdot R\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right)}{\beta \cdot P\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) \cdot R\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right)} \tag{20}$$

where β is 0.3, $P = \text{TP}/(\text{TP} + \text{FP})$, $R = \text{TP}/(\text{TP} + \text{FN})$, and TP, FP, FN are shown in detail as follows:

$$\begin{aligned} \text{TP}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) &= \sum_{j=1}^{h \times w} \hat{\boldsymbol{y}}^i(j) \cdot \boldsymbol{y}(j) \\ \text{FP}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) &= \sum_{j=1}^{h \times w} \hat{\boldsymbol{y}}^i(j) \cdot (1 - \boldsymbol{y}(j)) \\ \text{FN}\left(\hat{\boldsymbol{y}}^i, \boldsymbol{y}\right) &= \sum_{j=1}^{h \times w} \left(1 - \hat{\boldsymbol{y}}^i(j)\right) \cdot \boldsymbol{y}(j). \end{aligned} \tag{21}$$

In general, our loss function can be indicated as

$$\mathcal{L} = \mathcal{L}_{\text{bce}} + \lambda_1 \mathcal{L}_{\text{iou}} + \lambda_2 \mathcal{L}_{\text{fm}} + \lambda_3 \mathcal{L}_{\text{fg}}. \tag{22}$$

Finally, we use gradient descent to optimize our method, where the encoder f^i, the DSM g^i, and the decoder h^i are learnable and the others are fixed.

IV. Experiments

A. Experimental Protocol

1) Dataset: We evaluate our method and compared methods on three publicly available ORSI-SOD datasets to demonstrate the effectiveness of our method.

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1) *ORSSD* [22] contains 800 images and corresponding pixel-level annotations from previous RSI datasets and Google Earth, which includes a variety of challenging scenes such as aircraft, boats, cars, rivers, ponds, bridges, stadiums, and beaches. Typically, 600 images are used for training and 200 images are used for testing.
2) *EORSSD* [21] expands on the ORSSD dataset with more complex scenarios, resulting in 2000 images with well-annotated images, of which 1400 images are used for training and 600 images for testing.
3) *ORSI-4199* [25] is the latest and largest available ORSI-SOD dataset, with 4199 images with GT annotations, including nine challenging scenes, i.e., big salient object (BSO), small salient object (SSO), off-center (OC), complex salient object (CSO), complex scene (CS), narrow salient object (NSO), multiple salient objects (MSOs), low-contrast scene (LCS), and incomplete salient object (ISO). The dataset has 2000 train images and 2199 test images.

2) Evaluation Metrics: We select four widely used metrics to evaluate our method.

1) F-measure (F_β) [49] is the weighted harmonic average of precision and recall, where β is set to 0.3 to emphasize the precision over recall, and we adopt its maximum.
2) Structural measure (S_α) [50] is to evaluate the structural similarity of the saliency map and the corresponding GT at object-aware and region-aware levels, where α is set to 0.5 as suggested.
3) MAE evaluates the mean pixel-level difference between the saliency map and the corresponding GT.
4) Precision–recall curve (PR) plots different pairs of precision and recall values with the threshold ranging from 0 to 255.

3) Experimental Details: In our experiments, we implemented our proposed method using the PyTorch [51] platform, utilizing an NVIDIA 3090 GPU for accelerated computations. During the training of the network, each image is enhanced with flipped and rotated data augmentations and is resized to 224 × 224. We load the publicly available P2T-small model [31] as our pre-trained model. For network optimization, we use AdamW [52] optimizer and trained the network for 100 epochs. The initial learning rate was set to 10^{-4}, and we utilized the cosine annealing learning rate scheduler [53] to dynamically adjust the learning rate throughout the training process. Other hyperparameters, including weight decay (wd), deep supervision label weight coefficients $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, and loss function scaling factors $\{\lambda_1, \lambda_2, \lambda_3\}$, are determined based on the experimental results and tailored to the specific dataset being evaluated. The optimal hyperparameters for each dataset are presented in Table I, and a more comprehensive analysis of the hyperparameters is provided in Section IV-C1.

TABLE I
HYPERPARAMETERS CORRESPOND TO THE OPTIMAL PERFORMANCE ON THREE ORSI DATASETS, WHICH MAY VARY WITH THE DATASET

Dataset	wd	α_1	α_2	α_3	α_4	λ_1	λ_2	λ_3
ORSSD	0.1	1	0.8	0.2	0	2	1.2	1
EORSSD	0.001	1	0.8	0.2	0	2	1	1
ORSI4199	0.01	1	0.8	0.2	0	2	0.5	1

B. Comparison With State-of-the-Art Methods

We compare our method with 28 SOTA methods, which can be classified into five types. The first one is the traditional NSI-SOD, including RRWR [54], HDCT [55], DSG [56], SMD [57], and RCRR [58]. The second one is CNN-based NSI-SOD, including DSS [32], RADF [59], R3Net [60], EGNet [47], PoolNet [61], GCPA [62], ITSD [63], MINet [13], GateNet [14], U2Net [64], SUCA [65], and PA-KRN [66]. The third one is the traditional ORSI-SOD, including VOS [43], CMC [44], and SMFF [45]. The fourth one is CNN-based ORSI-SOD, including LVNet [22], DAFNet [21], EMFINet [23], SARNet [67], MJRBM [25], CorrNet [68], and FSMINet [69]. The last one is Transformer-based ORSI-SOD, including ASTT [20]. For a fair comparison, we use ORSI saliency maps from publicly available benchmarks or provided by the authors, and the NSI-SOD method is trained and tested with its default settings, using the same data as our method.

1) Quantitative Comparison: We report three important quantitative metrics, S_α, F_β, and MAE, for our method and 28 other comparison methods on the ORSSD and EORSSD datasets. We provide our method under both P2T [31] and Swin transformer (SwinT) [70] encoders, with the latter having a larger number of parameters and better performance. It can be clearly seen in Table II that our method exhibits excellent performance on both ORSI-SOD datasets compared to other comparative methods. On both datasets, our method (under SwinT) completely surpasses other comparative methods in all metrics. In particular, compared to the latest and also transformer-based ORSI-SOD ASTT [20] with us, even with the smaller encoder P2T, our method notably outperforms it on both datasets. Even when NSI-SODs are retrained on the ORSI dataset, their performance is still significantly lower than that of specially designed ORSI-SODs, which also illustrates the necessity of developing elaborate methods for ORSI.

Furthermore, we evaluate the PR curves of our method and 28 other advanced methods on two ORSI datasets. For better visualization, we highlight the better-performing methods. We have also zoomed in on two more intensive areas for a clearer comparison. Fig. 4 shows that our method guarantees both higher precision and recall. Specifically, when the recall is around 0.9, the precision of our method is ahead of all other comparison methods on both datasets. Overall, the PR curves of our method are closer to the upper right corner than the other methods, which is consistent with the superior quantitative results of our method in Table II.

2) Attributes-Based Performance on ORSI-4199: In the highly challenging ORSI-4199 dataset, each image represents typical challenging attributes encountered in remote-sensing scenarios. This dataset allows us to gain a better understanding of the strengths and weaknesses of different SOD methods. Table III shows the scores of our method and the comparison methods. It is evident that our method demonstrates superior performance overall, with our method being more

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TABLE II

QUANTITATIVE RESULTS FOR OUR METHOD AND OTHER COMPARATIVE METHODS ON TWO POPULAR ORSI DATASETS (I.E., ORSSD AND EORSSD) IN TERMS OF S_α, F_β, AND MAE METRICS. WE ALSO REPORT THE SPEED, NETWORK PARAMETERS (DENOTED BY PARA), AND FLOPS FOR ALL METHODS. THERE ARE 28 STATE-OF-THE-ART METHODS, INCLUDING FIVE TRADITIONAL NSI-SOD METHODS, 12 CNN-BASED NSI-SOD METHODS, THREE TRADITIONAL ORSI-SOD METHODS, SEVEN CNN-BASED ORSI-SOD METHODS, AND ONE TRANSFORMER-BASED ORSI-SOD METHOD. ↑ AND ↓ INDICATE THAT THE LARGER AND SMALLER THE CORRESPONDING METRICS ARE, RESPECTIVELY, THE BETTER. THE "−" SYMBOL INDICATES THAT THIS RESULT IS NOT AVAILABLE. THE BEST THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Type	Input Size	Speed (fps↑)	Para (M↓)	FLOPs (G↓)	ORSSD			EORSSD		
						$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
RRWR [54]	T.N.	-	0.3	-	-	0.6836	0.5590	0.1324	0.5999	0.4019	0.1677
HDCT [55]	T.N.	-	7	-	-	0.6197	0.5257	0.1309	0.5975	0.5431	0.1087
DSG [56]	T.N.	-	0.6	-	-	0.7196	0.6254	0.1041	0.6429	0.5259	0.1246
SMD [57]	T.N.	-	-	-	-	0.7642	0.6692	0.0715	0.7112	0.5915	0.0770
RCRR [58]	T.N.	-	0.3	-	-	0.6850	0.5591	0.1277	0.6014	0.4021	0.1644
DSS [32]	C.N.	400 × 300	8	62.23	114.6	0.8260	0.7467	0.0363	0.7879	0.6868	0.0186
RADF [59]	C.N.	400 × 400	7	62.54	214.2	0.8259	0.7619	0.0382	0.8195	0.7473	0.0168
R3Net [60]	C.N.	300 × 300	2	56.16	47.5	0.8143	0.7456	0.0399	0.8198	0.7516	0.0171
EGNet [47]	C.N.	380 × 320	9	108.07	291.9	0.8722	0.8333	0.0217	0.8606	0.7879	0.0109
PoolNet [61]	C.N.	400 × 300	25	53.63	123.4	0.8397	0.7705	0.0358	0.8221	0.7574	0.0206
GCPA [62]	C.N.	320 × 320	23	67.06	54.3	0.9019	0.8687	0.0171	0.8877	0.8346	0.0103
ITSD [63]	C.N.	288 × 288	16	17.08	54.5	0.9043	0.8735	0.0165	0.9045	0.8523	0.0106
MINet [13]	C.N.	320 × 320	12	47.56	146.3	0.9034	0.8760	0.0146	0.9035	0.8343	0.0093
GateNet [14]	C.N.	384 × 384	25	100.02	108.3	0.9193	0.8871	0.0137	0.9109	0.8566	0.0095
U2Net [64]	C.N.	-	25	44.01	376.2	0.9156	0.8739	0.0166	0.9209	0.8732	0.0076
SUCA [65]	C.N.	256 × 256	24	117.71	56.4	0.8990	0.8484	0.0145	0.8992	0.8229	0.0097
PA-KRN [66]	C.N.	600 × 600	16	141.06	617.7	0.9231	0.8890	0.0139	0.9185	0.8640	0.0104
VOS [43]	T.O.	-	-	-	-	0.5366	0.3471	0.2151	0.5083	0.2776	0.2096
CMC [44]	T.O.	-	-	-	-	0.6036	0.3913	0.1267	0.5798	0.3272	0.1057
SMFF [45]	T.O.	-	-	-	-	0.5313	0.4417	0.1854	0.5407	0.5207	0.1434
LVNet [22]	C.O.	128 × 128	1.4	-	-	0.8813	0.8263	0.0207	0.8650	0.7824	0.0145
DAFNet [21]	C.O.	128 × 128	26	29.35	68.5	0.9192	0.8928	0.0113	0.9176	0.8612	0.0060
EMFINet [23]	C.O.	256 × 256	25	107.26	480.9	0.9365	0.9002	0.0109	0.9299	0.8720	0.0084
SARNet [67]	C.O.	336 × 336	47	25.91	129.7	0.9138	0.8851	0.0189	0.9249	0.8720	0.0100
MJRBM [25]	C.O.	352 × 352	32	43.54	95.7	0.9194	0.8842	0.0163	0.9200	0.8656	0.0099
CorrNet [68]	C.O.	256 × 256	100	4.09	21.09	0.9392	0.9129	0.0098	0.9298	0.8778	0.0083
FSMINet [69]	C.O.	384 × 384	28	3.39	11.02	0.9355	0.9041	0.0101	0.9356	0.8678	0.0079
ASTT [20]	Tf.O.	224 × 224	13	43.12	23.35	0.9345	0.9061	0.0094	0.9250	0.8740	0.0059
ours (P2T)	Tf.O.	224 × 224	48	31.64	4.87	0.9422	0.9158	0.0084	0.9314	0.8752	0.0055
ours (SwinT)	Tf.O.	224 × 224	38	116.51	27.26	0.9446	0.9174	0.0077	0.9328	0.8783	0.0052

T.N.: Traditional NSI-SOD method; C.N.: CNN-based NSI-SOD method; T.O.: Traditional RSI-OSOD method; C.O.: CNN-based ORSI-SOD method; Tf.O.: Transformer-based ORSI-SOD method.

advantageous in two-thirds of the scenarios (BSO, CS, CSO, ISO, NSO, and OC), while MJRBM is more advantageous in the remaining third of the scenarios (LCS, MSO, and SSO). Notably, our method ranks first in terms of a higher average score than other compared methods. This result illustrates that our method remains outstanding in challenging scenarios.

3) Qualitative Comparison: We show in Fig. 5 the visualization results from the EORSSD dataset under some challenging scenarios, including BSO, multiple and tiny salient objects, NSO, low-contrast salient object, and OC salient object. We provide salient maps of representative models in five categories to provide a clear comparison. It is evident that traditional methods based on handcrafted features exhibit a significant gap compared to deep learning methods. These traditional methods struggle to ensure the integrity of salient objects and the clarity of edge details. Additionally, the CNN-based NSI-SOD method exhibits unsatisfactory results compared to the well-designed ORSI-SOD method, especially the results are very poor when facing the low-contrast salient object and OC salient object (see results of PR-KRN in Fig. 5(d) and (e) group scenes). ORSI-SOD methods generally produce better saliency maps compared to other methods.

TABLE III

ATTRIBUTE-BASED PERFORMANCE ON THE CHALLENGING DATASET ORSI4199. SIMILAR TO [25], THESE METHODS ARE TRAINED ON THE ORSI4199 TRAINING SET AND TESTED UNDER EACH OF THE NINE CHALLENGING SCENARIOS DIVIDED BY THE TESTING SET. WE REPORT F_β METRICS UNDER THE NINE SCENARIOS ALONG WITH THEIR AVERAGE VALUE AS [25]. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Attr	NLDF CVPR17 [71]	PFAN CVPR19 [72]	RAS TIP20 [73]	MJRBM TGRS21 [25]	HFANet TGRS22 [27]	ASTT TGRS23 [20]	ours
BSO	0.8808	0.8969	0.8209	0.9166	0.9188	0.9180	**0.9296**
CS	0.8255	0.8615	0.8132	0.8961	0.8956	0.9085	**0.9191**
CSO	0.8214	0.8568	0.8106	0.8899	0.8862	0.8901	**0.9013**
ISO	0.8600	0.8739	0.8179	0.8937	0.9028	0.9226	**0.9236**
LCS	0.7115	0.7620	0.7358	**0.7916**	0.7687	0.7847	0.7910
MSO	0.7736	0.8064	0.8142	**0.8585**	0.8473	0.8564	0.8565
NSO	0.6796	0.8442	0.6904	0.8748	0.8846	0.8976	**0.9138**
OC	0.7151	0.7675	0.7642	0.8422	0.8426	0.8368	**0.8450**
SSO	0.6921	0.7338	0.7573	**0.8091**	0.7878	0.7999	0.7931
Avg	0.7733	0.8226	0.7805	0.8636	0.8594	0.8683	**0.8748**

However, they still face challenges in accurately capturing complete salient objects or preserving clear edge details in certain scenarios. This limitation arises from their difficulty in

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effectively learning both deep semantic features and shallow detail features simultaneously. For example, ASTT may excel in capturing edge details but struggle to predict the salient objects in their entirety (see the third and fifth rows in Fig. 5) or incorrectly predict the background as a salient object (see the first and ninth rows in Fig. 5). DAFNet, on the other hand, ensures the integrity of salient targets in most cases but may produce rough edges (see first and second rows in Fig. 5). In contrast, our ensemble learning method effectively learns multigranularity simultaneously. This allows us to guarantee both the object integrity and the edge detail in these challenging scenarios.

C. Ablation Studies

In this section, we maintain consistent settings across all experiments, except for the variables being examined. Furthermore, we report the best experimental result obtained after 100 epochs to showcase the validity of our proposed approach. It is important to note that the ablation studies used P2T as the encoder, except for examining the impact of different encoders.

1) Hyperparametric Analysis: In our method, there are two main aspects of hyperparameters: those related to the optimizer (such as learning rate and weight decay) and those related to the method itself (such as deep supervision label weight coefficients $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ and loss function scaling factors $\{\lambda_1, \lambda_2, \lambda_3\}$). Table I presents the hyperparameters that yield relatively better performance based on our experimental results. The learning rate is set to 1e-4, and the weight decay, deep supervision label weight coefficients, and loss function scaling factors are determined based on the specific dataset. Table IV demonstrates the effect of weight decay on different datasets. A weight decay setting of 0.1 is found to be more suitable for ORSSD, 0.001 for EORSSD, and 0.01 for ORSI-4199. These settings are determined based on the preference for F_β when performance is similar. We also investigate the impact of deep supervision label weights and loss function scaling factors on the ORSSD dataset. Fig. 6(a) justifies that we set $\{\alpha_1, \alpha_4\}$ to $\{1, 0\}$. Regarding the hyperparameters $\{\alpha_2, \alpha_3\}$ for the mixed features, the setting $\{0.8, 0.2\}$ is more suitable for the ORSSD, as shown in Fig. 6(b). Table V shows that different loss function scaling factors do have an impact on performance, and $\{2, 1.2, 1\}$ yield relatively better results for ORSSD. Similarly, we determine hyperparameters with relatively good performance for the EORSSD and ORSI-4199 datasets based on our experimental results.

2) Impact of Different Encoders: In this section, we explore the effectiveness of our ensemble learning method under different encoders. Firstly, we present the experimental results of our method using five encoders in two distinct structures: three transformer-based encoders ("P2T-small [31]," "PVT-v2 [74]," and "SwinT-base [70]") and two CNN encoders ("ResNet34 [75]" and "ResNet50 [75].") As shown in Table VI, the three transformer-based encoders with global receptive fields exhibit superior performance, and larger "SwinT-base" performs better. In contrast, the two CNN encoders perform poorly due to the shallow receptive fields are too small to guarantee the weak learner in the shallow branches which is necessary for ensemble learning, and even larger "ResNet50" performs lower. After a comprehensive comparison, we choose P2T and SwinT as the encoder of our method. Concretely, for the transformer-based P2T encoder, we provide two variants: "P2T baseline" and "SwinT baseline" (comprising solely the encoder and decoder). By integrating our IDEL (DSM, hardness-aware weights, and fusion) into the baseline, we obtain "P2T baseline + IDEL" and "SwinT baseline + IDEL." As shown in Table VII, with an almost negligible increase in network parameters, our method achieves a dramatic performance improvement across three metrics on two datasets for both P2T and SwinT encoders. Notably, MAE is reduced by 20.0%–32.1% for P2T, and by 28.7%–29.7% for SwinT, compared to the corresponding baseline on ORSSD and EORSSD datasets. Subsequently, we give an experimental validation when facing the SOTA CNN-based method MCCNet [24], and we add our hardness-aware weights and fusion to "MCCNet" to obtain "MCCNet + IDEL." Despite the CNN-based encoder not meeting the necessary conditions for ensemble learning (due to a small shallow perceptual field that may not ensure weak learners in the shallow branches), we still achieve performance improvement by incorporating our IDEL with the same network parameters and FLOPs. However, considering the large computational size and poor interpretability of "MCCNet + IDEL," it was not our primary choice. Overall, our IDEL produces superior results using both CNN-based and transformer-based encoders, with minimal increase in network parameters and FLOPs. This is attributed to the compact DSM and the use of hardness-aware weights without backpropagation gradients, which contribute to the effectiveness of our approach.

3) Impact of the Deep Supervision Module: To further validate the effectiveness of the DSM, we design five variants for comparison. The "w/o DSM" variant indicates the removal of the DSM based on our method, and the "w/ CRM" variant replaces the DSM with a hybrid structure of Transformer and convolutional block similar to CRM [42]. The "w/ Inception" variant replaces the DSM with a structure similar to Inception in GoogleNet [76]. The "w/ Maxpool" variant replaces the spatial attention in the DSM with Maxpool, while the "w/ Avgpool" variant replaces the spatial attention with Avgpool. As shown in Table VIII, the removal of DSM shows a different degree of decrease in all three metrics. The "w/ CRM" variant is slightly higher than our method in S_α, but lower than our method in the remaining two metrics and brings larger network parameters and FLOPs. The "w/ Inception" variant is lower than our method in all metrics. This demonstrates that our elaborate DSM achieves performance gains at the cost of only a slight increase in network parameters and FLOPs. The "w/ Maxpool" and "w/ Avgpool" exhibit a decrease in performance across all three metrics, which confirms the effectiveness of the spatial attention in the DSM.

4) Effect of Hardness-Aware Weights and Fusion: In this part, we explore the effects of hardness-aware weights and fusion. To demonstrate their impact, we remove hardness-aware weights, fusion, and both hardness-aware weights and fusion from our method. These modifications are denoted as "w/o HW" (without hardness-aware weights), "w/o F" (without fusion), and "w/o HW and F" (without both

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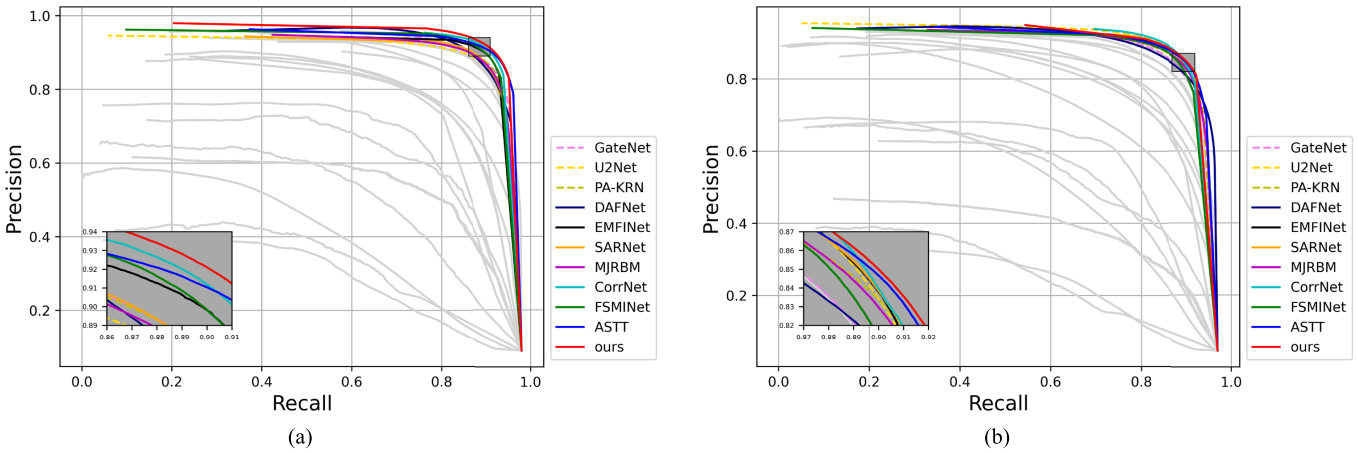

Fig. 4. PR curves of our method and other compared methods on two datasets, i.e., (a) ORSSD and (b) EORSSD. We highlight some of the methods with better performance for a clear comparison. Zoom in for better view.

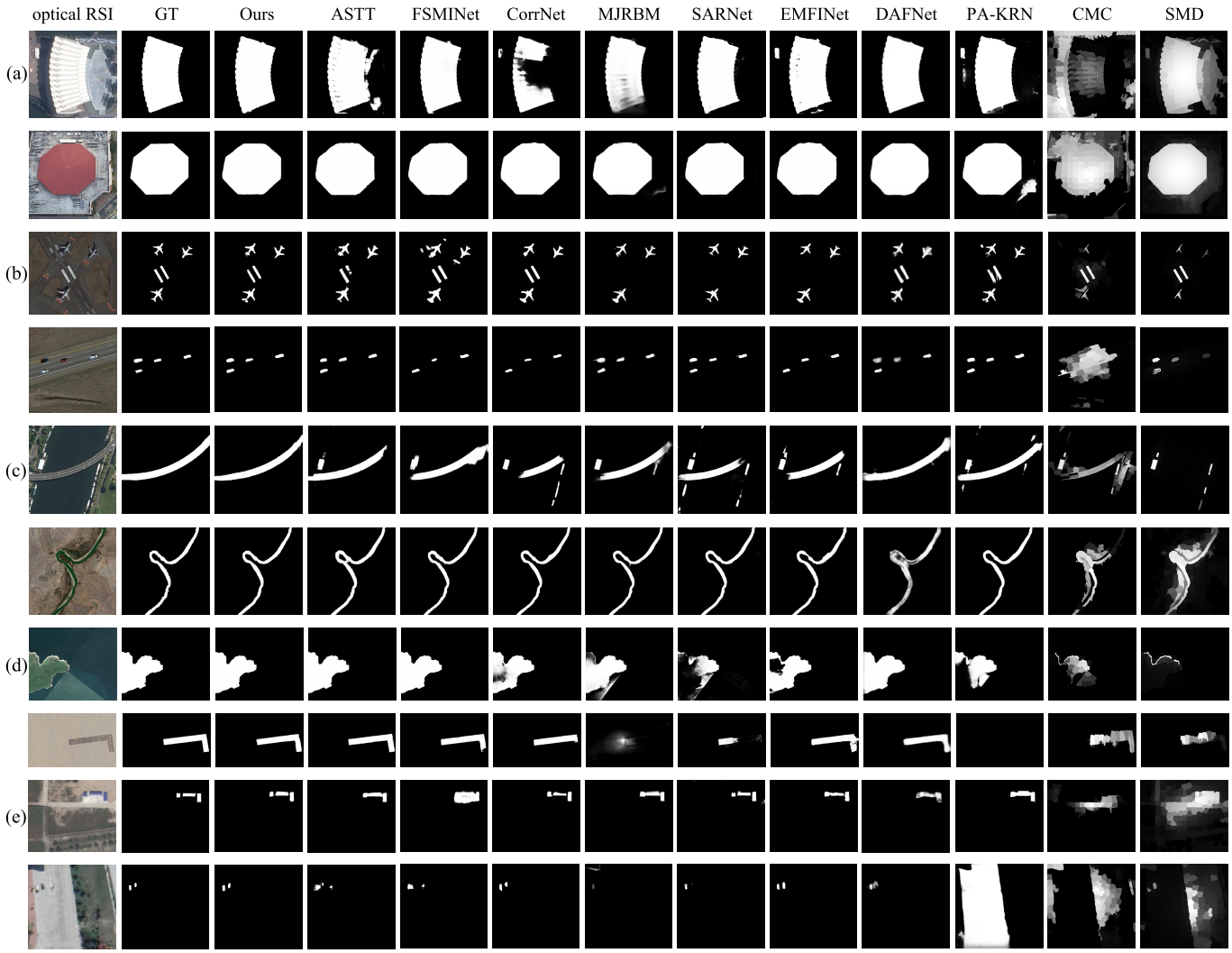

Fig. 5. Saliency maps comparison with ten representative methods on the EORSSD testing set. We select several challenging scenarios, including (a) BSO, (b) multiple and tiny salient objects, (c) NSO, (d) low-contrast salient object, and (e) OC salient object. Zoom in for better view.

hardness-aware weights and fusion), respectively. Table IX shows that the modified models all show different degrees of performance degradation, which proves the effectiveness of hardness-aware weights and fusion. It is evident that hardness-aware weights and fusion are inseparable and contribute to significant performance improvements. This finding

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TABLE IV

COMPARISON RESULTS OF THE THREE ORSI DATASETS UNDER DIFFERENT WEIGHT DECAY IN TERMS OF S_α, F_β, AND MAE METRICS. THE BEST RESULT OF EACH COLUMN IS **BOLD**

wd	ORSSD			EORSSD			ORSI4199		
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
0.1	0.9422	**0.9158**	**0.0084**	0.9307	0.8729	0.0066	0.8765	0.8671	0.0313
0.01	**0.9430**	0.9151	**0.0084**	0.9241	0.8714	0.0062	**0.8786**	0.8678	**0.0290**
0.001	0.9419	0.9101	0.0091	**0.9314**	**0.8752**	**0.0055**	0.8682	**0.8682**	0.0320

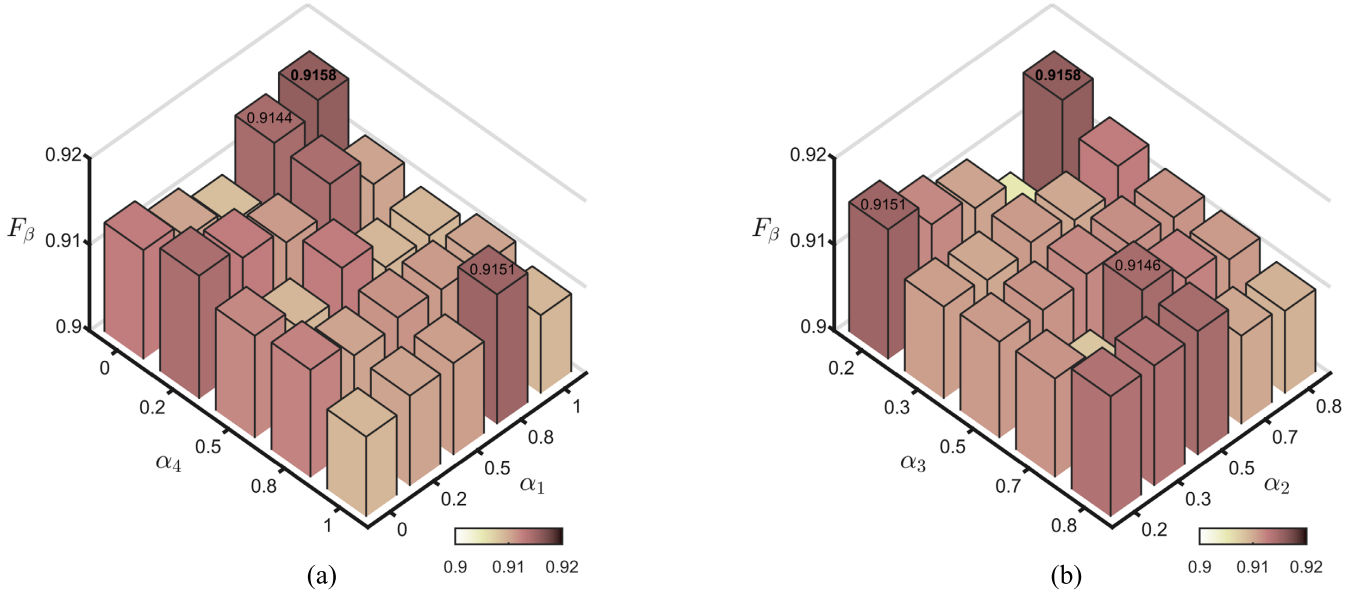

Fig. 6. Metric F_β with different deep supervised labeling weight coefficients $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ on ORSSD, i.e., (a) $\{\alpha_1, \alpha_4\}$ and (b) $\{\alpha_2, \alpha_3\}$. We highlight the top three F_β metrics and the best is bold. Zoom in for better view.

TABLE V

COMPARISON RESULTS UNDER DIFFERENT LOSS FUNCTION SCALING FACTORS $\{\lambda_1, \lambda_2, \lambda_3\}$ IN TERMS OF S_α, F_β, AND MAE METRICS ON ORSSD. THE BEST RESULT OF EACH COLUMN IS **BOLD**

λ_1	λ_2	λ_3	ORSSD		
			$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
0.5	1	1	0.9401	0.9097	0.0092
1	0.5	1	0.9396	0.9075	0.0094
1	1	0.5	0.9386	0.9094	0.0089
1	1	1	0.9385	0.9098	0.0096
1	1	2	0.9368	0.9051	0.0096
1	2	1	0.9394	0.9076	0.0099
2	1	1	0.9408	0.9113	**0.0084**
2	1.5	1	0.9420	0.9119	0.0096
2	1.2	1	**0.9422**	**0.9158**	**0.0084**

TABLE VI

EXPERIMENTAL RESULTS EXPLORING THE EFFECTIVENESS OF OUR ENSEMBLE LEARNING METHOD UNDER DIFFERENT ENCODERS ON ORSSD AND EORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Encoder	Input Size	Para	FLOPs	ORSSD		
				$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
P2T-small	224×224	31.55	4.87	0.9422	0.9158	0.0084
PVT-v2	224×224	32.51	5.02	0.9440	0.9158	0.0092
SwinT-base	224×224	116.51	27.26	**0.9446**	**0.9174**	**0.0077**
ResNet34	224×224	28.76	74.46	0.9363	0.9096	0.0112
ResNet50	224×224	42.83	108.81	0.9326	0.9061	0.0115

aligns with our idea of ensemble learning. To further illustrate the effects of hardness-aware weights and fusion, we also provide visualization results of different branch saliency maps $\{\hat{\boldsymbol{y}}^1, \hat{\boldsymbol{y}}^2, \hat{\boldsymbol{y}}^3, \hat{\boldsymbol{y}}^4\}$ and final saliency map $\hat{\boldsymbol{y}}$ before and after removing hardness-aware weights and fusion. As can be seen from Fig. 7, our method is more accurate in predicting difficult pixels near the edges due to the presence of hardness-aware weights. Although the saliency maps of different branches of our method have some flaws, such as missing pixels inside the saliency map $\hat{\boldsymbol{y}}^1$, or rough edges in $\hat{\boldsymbol{y}}^4$, fusion effectively combines the results from different branches to obtain a satisfactory final saliency map $\hat{\boldsymbol{y}}$. This highlights the importance of hardness-aware weights and fusion in ensuring the object integrity and the edge detail.

5) Effect of Different Loss Functions: Our method employs a hybrid loss function including multiple loss functions of BCE loss, IoU loss, Fm loss, and Fg loss. Considering that the BCE loss function is the most commonly used in ORSI-SOD, in this section, we remove the IoU, Fm, and Fg loss functions, respectively, to verify the effectiveness of the hybrid

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TABLE VII

RESULTS OF ABLATION STUDY EXPLORING THE VALIDITY OF OUR ENSEMBLE LEARNING METHOD UNDER DIFFERENT ENCODERS ON ORSSD AND EORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. THE BETTER RESULT OF EACH ENCODER IS **BOLD**

Models	Encoder	Input Size	Para	FLOPs	ORSSD			EORSSD		
					$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
P2T baseline	P2T-small	224×224	31.55	4.87	0.9329	0.9064	0.0105	0.9088	0.8641	0.0081
P2T baseline+IDEL	P2T-small	224×224	31.64	4.87	**0.9422**	**0.9158**	**0.0084**	**0.9314**	**0.8752**	**0.0055**
SwinT baseline	SwinT-base	224×224	116.13	27.26	0.9355	0.9112	0.0108	0.9213	0.8671	0.0074
SwinT baseline+IDEL	SwinT-base	224×224	116.51	27.26	**0.9446**	**0.9174**	**0.0077**	**0.9328**	**0.8783**	**0.0052**
MCCNet	VGG-16	256×256	64.52	109.1	0.9437	0.9155	**0.0087**	0.9327	0.8904	0.0066
MCCNet+IDEL	VGG-16	256×256	64.52	109.1	**0.9453**	**0.9176**	0.0094	**0.9396**	**0.8933**	**0.0061**

TABLE VIII

RESULTS OF ABLATION EXPERIMENTS TO EVALUATE THE CONTRIBUTION OF DSM. COMPARISON OF OUR METHOD AND THE FIVE VARIANTS FOR DSM ON ORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Variants	Para (M↓)	FLOPs (G↓)	ORSSD		
			$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
w/o DSM	31.55	4.87	0.9413	0.9118	0.0094
w/ CRM	38.23	5.83	**0.9432**	0.9121	0.0094
w/ Inception	31.64	4.90	0.9386	0.9083	0.0092
w/ Maxpool	31.64	4.87	0.9420	0.9121	0.0094
w/ Avgpool	31.64	4.87	0.9364	0.9062	0.0095
ours	31.64	4.87	0.9422	**0.9158**	**0.0084**

TABLE IX

RESULTS OF ABLATION EXPERIMENTS EVALUATING THE EFFECT OF HARDNESS-AWARE WEIGHTS AND FUSION. THE COMPARISON RESULTS ARE CONSTRUCTED ON ORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. "W/O" INDICATES THE RESPECTIVE PART REMOVED BASED ON OUR METHOD. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Variants	Components		ORSSD		
	weight	fusion	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
ours	✓	✓	**0.9422**	**0.9158**	**0.0084**
w/o HW		✓	0.9379	0.9101	0.0086
w/o F	✓		0.9343	0.9101	0.0146
w/o HW and F			0.9389	0.9092	0.0110

TABLE X

RESULTS OF ABLATION EXPERIMENTS TO EVALUATE THE EFFECTIVENESS OF OUR HYBRID LOSS FUNCTION. COMPARISON OF OUR METHOD AND THREE VARIANTS FOR LOSS FUNCTION ON ORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Variants	Components				ORSSD		
	bce	iou	fm	fg	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
ours	✓	✓	✓	✓	**0.9422**	**0.9158**	**0.0084**
w/o iou	✓		✓	✓	0.9386	0.9102	0.0102
w/o fm	✓	✓		✓	0.9416	0.9120	0.0092
w/o fg	✓	✓	✓		0.9414	0.9120	0.0086

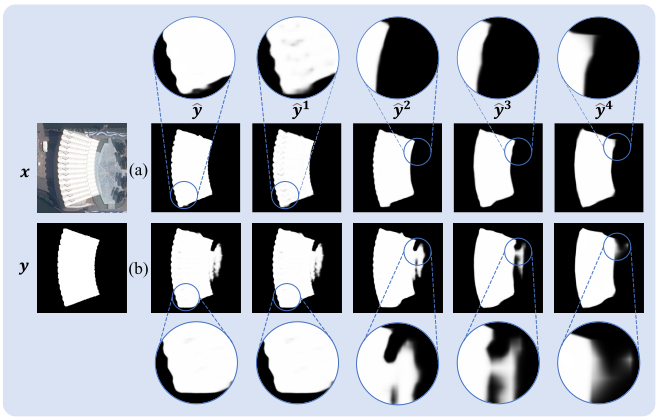

Fig. 7. Comparison of visualization results of different branch saliency maps and the final saliency map before and after removing the hardness-aware weights and fusion, (a) corresponds to "ours" and (b) corresponds to "w/o HW and F." In (b), we treat $\hat{y}^1$ as the final saliency map $\hat{y}$.

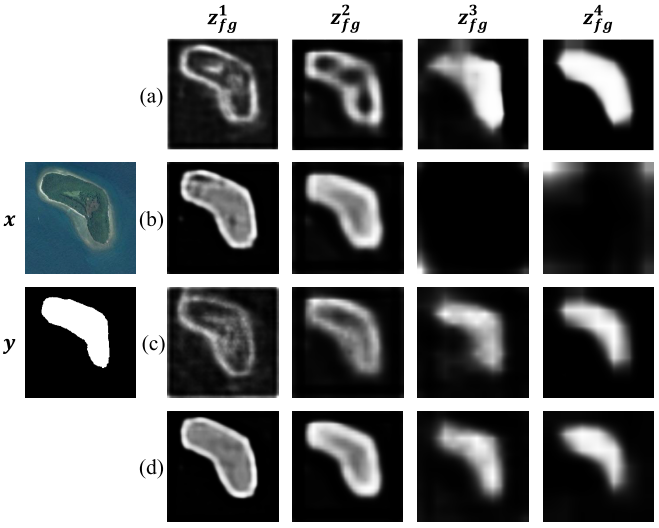

Fig. 8. Comparison of foreground feature visualization results for different branches when using different deep supervision labels. (a) Corresponds to "DSL-a." (b) Corresponds to "DSL-b." (c) Corresponds to "DSL-c." (d) Corresponds to "ours."

loss function, which are denoted by "w/o iou," "w/o fm," and "w/o fg," respectively. From Table X, we observe that the performance of the "w/o iou" variant drops more, indicating that the IoU loss function is more important to our method and we can also see that the IoU loss function occupies a higher percentage in Table V, which is consistent with both. The "w/o fm" and "w/o fg" variants show degradation in performance on all metrics. This suggests that both the Fm and Fg loss functions contribute to the overall effectiveness of our method. Overall, the hybrid loss function we employ proves to be efficient, allowing us to achieve performance gains without increasing the network parameters and FLOPs.

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TABLE XI

ANALYTICAL RESULTS ON DEEP SUPERVISION LABELS. COMPARISON OF OUR METHOD AND THREE VARIANTS FOR DEEP SUPERVISION LABELS ON ORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. FOR DETAILS ON detail AND body SEE FIG. 2. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Variants	$\boldsymbol{y}^1_{ds}$	$\boldsymbol{y}^2_{ds}$	$\boldsymbol{y}^3_{ds}$	$\boldsymbol{y}^4_{ds}$	ORSSD		
					$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
DSL-a	$\boldsymbol{y}^1_{dt}$	$\boldsymbol{y}^1_{dt}$	$\boldsymbol{y}^1_{dt}$	$\boldsymbol{y}^1_{dt}$	0.9435	0.9116	0.0087
DSL-b	$\boldsymbol{y}^1_{dt}$	$0.8 \times \boldsymbol{y}^1_{dt} + 0.2 \times \boldsymbol{y}^1_{bd}$	$0.2 \times \boldsymbol{y}^1_{dt} + 0.8 \times \boldsymbol{y}^1_{bd}$	$\boldsymbol{y}^1_{bd}$	0.9412	0.9112	0.0086
DSL-c	$\boldsymbol{y}^2_{dt}$	$0.8 \times \boldsymbol{y}^2_{dt} + 0.2 \times \boldsymbol{y}^2_{bd}$	$0.2 \times \boldsymbol{y}^2_{dt} + 0.8 \times \boldsymbol{y}^2_{bd}$	$\boldsymbol{y}^2_{bd}$	**0.9443**	0.9138	**0.0084**
ours	$\boldsymbol{y}^1_{dt}$	$0.8 \times \boldsymbol{y}^1_{dt} + 0.2 \times \boldsymbol{y}^1_{bd}$	$0.2 \times \boldsymbol{y}^2_{dt} + 0.8 \times \boldsymbol{y}^2_{bd}$	$\boldsymbol{y}^2_{bd}$	0.9422	**0.9158**	**0.0084**

TABLE XII

RESULTS OF ABLATION EXPERIMENTS TO EVALUATE THE EFFECTIVENESS OF OUR DEEP SUPERVISION LOSS FUNCTION. COMPARISON OF OUR METHOD AND TWO VARIANTS FOR DIFFERENT DEPTH SUPERVISION LOSS FUNCTIONS ON ORSSD IN TERMS OF S_α, F_β, AND MAE METRICS. THE BEST RESULT OF EACH COLUMN IS **BOLD**

Variants	ORSSD		
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$MAE \downarrow$
w/ L1	0.9406	0.9086	**0.0081**
w/ BCE	**0.9432**	0.9132	0.0086
ours	0.9422	**0.9158**	0.0084

6) Analysis of Deep Supervision Labels: In this section, we analyze the deep supervision labels used in our method and provide three variants for comparison. The "DSL-a" variant represents the most common approach in OSRI-SOD, where all branches use edge information as deep supervision labels. The "DSL-b" variant adopts the first label decoupling method shown in Fig. 2, treating the sum of the weights of the two parts as the deep supervision labels for different branches. The "DSL-c" variant utilizes the second label decoupling method shown in Fig. 2, treating the sum of the weights of the two parts as the deep supervision labels for different branches. We also provide visualization results for the foreground features $\{z^1_{\text{fg}}, z^2_{\text{fg}}, z^3_{\text{fg}}, z^4_{\text{fg}}\}$ of our method and three variants, and the foreground features are exactly the features that need to be supervised by deep supervision labels. From Table XI and Fig. 8, we observe that the "DSL-a" variant performs poorly due to the mismatch between the inductive biases of different branches when using edge information as deep supervision labels for all branches. The "DSL-b" variant exhibits poor performance because its third and fourth branches have inadequate foreground features, despite the good foreground features in the first and second branches. The "DSL-c" variant has a better performance and good foreground features for each branch. However, we notice that the foreground features of the first and second branches of the "DSL-c" variant are not as good as those of the "DSL-b" variant. Therefore, we combine the "DSL-b" and "DSL-c" variants to achieve both good performance and foreground features, which become the final solution for our deep supervision labels design. Additionally, we adopt the MSE loss function for deep supervision. Here, we consider two other variants, "w/ L1" replaces the MSE loss function with the $L1$ loss function, and "w/ BCE" replaces the MSE loss function with the BCE loss function. After comprehensive evaluation based on the three metrics, we select the MSE loss function as the most suitable choice (see Table XII).

V. CONCLUSION

Based on the performance results of our proposed novel IDEL framework for ORSI-SOD, we observe that our IDEL effectively addresses the challenge of learning features at multiple granularities. Within IDEL, the incorporation of Transformer blocks with a global receptive field and the DSM ensures the performance of each branch. The implementation of hierarchical deep supervision labels and hardness-aware pixel-wise weights enables the branches to learn distinct features. Moreover, the fusion module fully combines all outputs to generate the final saliency maps. Through extensive experiments and comparisons with 28 SOTA methods on three publicly available ORSI datasets, we have demonstrated the superiority of our approach. The minimal network parameters and computational complexity of our method render it suitable for practical applications, such as acquiring some important targets in scenes and subsequently performing classification, segmentation, and so on. At present, IDEL still needs high-quality annotations to ensure model performance. Thus, our future direction is to explore how to train high-performance models under weak annotations. Overall, our IDEL framework provides a novel and effective solution for ORSI-SOD, and we believe it has the potential to advance the field further.

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